

Designing AI Coaching Support for ADHD Family Co-Regulation

LUCAS M. SILVA, University of Iowa, USA

Children with Attention-Deficit/Hyperactivity Disorder (ADHD) face heightened challenges with self-regulation. While family members can support ADHD members via co-regulation, many wish for and need guidance on how to best provide support. Additionally, ADHD families face cyclical, whole-family self-regulation challenges further exacerbate challenges. In my an ongoing research program, I investigate how family-centered systems might help ADHD families gain awareness about individual struggles and provide mutual caregiving. I am now investigating how AI coaching can responsibly support family co-regulation: helping parents apply evidence-based strategies and helping families make sense of shared behavioral data together. Two ongoing design probe development anchor this work: an AI-coaching evaluation platform that lets clinicians and parents assess and interact with simulated coaching dialogues before any AI reaches families, and a working prototype of an in-home AI companion for parent- and family-facing co-regulation and discussion guidance, intended for future deployment with families. Drawing on this trajectory, I raise questions about family-centered AI being clinically grounded, reflect families' design preferences, and safely service members of different ages and literacy levels – questions I hope to develop with the workshop community around defining, evaluating, and governing family-centered AI.

1 Motivation and Position

Children with Attention-Deficit/Hyperactivity Disorder (ADHD) can face heightened challenges with self-regulation: managing emotions, sustaining goal-directed behavior, and adapting to everyday demands [1, 2]. These difficulties are frequently misread as willful non-compliance [3, 4] rather than genuine struggles with impulse control and attention, and, left unsupported, they can compound into academic, health, and lifelong difficulties [5, 6]. Parents, siblings, and other close family members can mitigate these challenges through *co-regulation*: the collaborative process by which others help children recognize, manage, and recover from dysregulation through emotional scaffolding, goal-setting, redirection, and role modeling [7, 8].

Co-regulation in ADHD families is rarely one-directional. Parents are often navigating their own self-regulation and mental health alongside their child's [9], and a child's dysregulation can elevate parental stress in ways that further erode the family's capacity to co-regulate [10]. This self-reinforcing cycle is a relational, whole-family dynamic captured via Family Systems lens. ADHD family challenges are not only attributes of an individual child but properties of an interconnected family unit, which motivates my position that family-centered AI for these contexts must be designed and evaluated at the level of the family system, not the individual user.

My prior work has shown that shared tracking of moods, goals, and daily behaviors across smartwatches and home displays can support children's self-monitoring [11], mediate family awareness of regulatory challenges [12], and foster mutual caregiving through collaborative data sharing [13, 14]. Across these studies, a consistent limitation emerges: families report wanting more than shared awareness of one another's struggles. They want scaffolding to interpret their data, and concrete, contextual guidance on how to act – needs that motivate my current project and that I believe are broadly relevant to how the workshop community defines and designs family-centered AI.

2 Toward Clinically-Grounded AI Coaching for Family Co-Regulation

My current project investigates how AI-mediated coaching can responsibly support, rather than undermine, co-regulation in ADHD families across everyday family contexts. At a high level, the work targets two related needs identified in my prior studies: parent-facing guidance for applying evidence-based co-regulation and self-regulation

strategies, and AI-facilitated support for family-time conversation and joint sensemaking around shared behavioral data. Two design probes from this work-in-progress speak directly to this workshop's themes of defining family-centered AI, understanding intergenerational dynamics, and building governance mechanisms for AI in the home.

2.1 An AI-Coaching Evaluation Platform: Grounding Before Deployment

A central design commitment of this project is that AI guidance should not reach families until it has been evaluated, iterated, and validated by the clinicians and parents. Involving these stakeholders in judging AI's clinical soundness and family appropriateness is necessary to mitigate risks and drive use for families' real interests. I have built a web-based evaluation platform, powered by large language models grounded in evidence-based frameworks such as Parent Management Training and healthy-conversation facilitation, that lets clinical experts and parents review simulated coaching dialogues drawn from de-identified prior family tracking data and interact with the coach directly to probe its responses. Participants rate dimensions such as clinical soundness, tone, safety signals, and child autonomy; and separately evaluate how well the system can facilitate joint family discussion around shared data. I see treating clinicians and parents as active evaluators and co-designers of the AI's behavior, rather than passive recipients, is one component to a larger answer for the workshop's question on "how should families and their care networks retain meaningful authority over an AI system introduced into sensitive, high-stakes home interactions?" Questions remain on involving children in a similar way at such a stage while maintaining safety and allowing for exploration and evaluation.

2.2 A Prototype AI Companion for Family Co-Regulation

Alongside the AI evaluation platform, I have developed a working prototype of an in-home AI companion intended to offer parent- and family-facing co-regulation support and discussion guidance, integrating conversational coaching with a home display and passively and manually collected family data. The prototype is designed to be redesigned: it currently serves as a proof-of-concept probe whose interaction patterns, tone, and scope of authority are meant to be reshaped by findings from the clinical and parent evaluations described above, ahead of a planned longitudinal deployment with families. I am interested in exploring ways that a staged approach (e.g., probe, evaluate, refine, and then deploy) as one potential path to allowing continuous AI governance in families.

3 Open Questions for the Workshop

Building on this trajectory, I bring the following questions to the workshop's activities:

- **Defining the family unit in AI coaching.** My work currently differentiates parent-facing and family-facing guidance, but ADHD families often include siblings, extended caregivers, and school or clinical partners whose needs and authority differ. How should family-centered AI represent and negotiate guidance across such asymmetric, multi-stakeholder units, especially for neurodivergent children whose developmental needs vary widely by age and ability?
- **Whose evaluation should count, and how.** Clinical experts and parents can disagree on what counts as sound or appropriate coaching. How should family-centered AI design processes weigh and reconcile clinical, parental, and (eventually) child perspectives when they diverge, particularly before a system is trusted with unsupervised interaction in the home?
- **Operationalizing risk mitigation and research ethics.** My evaluation-before-deployment pipeline is one attempt to predict and avoid risks to children's safety before an AI system reaches families. What safety and

ethical constraints should govern this kind of research, and what precautions, IRB considerations, and consent and disclosure practices are needed to responsibly study AI with families?

I look forward to discussing these questions and my research the workshop community, and to contributing to a shared research and design agenda for family-centered AI.

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